The Chain Rule

ANSWER KEY

Basic chain rule applications

1 Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$
$= 5(4x - 1)^4(4) = 20(4x - 1)^4$

b $f(x) = \sin(3x^2)$ $= 6x\cos(3x^2)$

c $f(x) = e^{5x}$ $= 5e^{5x}$

d $f(x) = \ln(2x + 7)$ $= \frac{2}{2x + 7}$

2 Differentiate $f(x) = \sqrt{x^2 + 9}$.
$f'(x) = \frac{2x}{2\sqrt{x^2 + 9}} = \frac{x}{\sqrt{x^2 + 9}}$.

The chain rule with trig and exponential functions

3 Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).
$f'(x) = 3\cos^2 x \cdot (-\sin x) = -3\cos^2 x \sin x$.

4 Differentiate $f(x) = e^{\sin x}$.
$f'(x) = \cos x \cdot e^{\sin x}$.

5 Differentiate $f(x) = \tan(e^x)$.
$f'(x) = \sec^2(e^x) \cdot e^x$.

Nested (multiple) chain rule

6 Differentiate $f(x) = \sin\left(\sqrt{x^2 + 1}\right)$, applying the chain rule twice.
$f'(x) = \cos\left(\sqrt{x^2 + 1}\right) \cdot \frac{x}{\sqrt{x^2 + 1}}$.

7 Differentiate $f(x) = (\ln(3x))^4$.
$f'(x) = 4(\ln(3x))^3 \cdot \frac{3}{3x} = \frac{4(\ln(3x))^3}{x}$.

Chain rule combined with product/quotient rules

8 Differentiate $f(x) = x^2 e^{3x}$.
$f'(x) = 2xe^{3x} + x^2(3e^{3x}) = e^{3x}(2x + 3x^2)$.

9 Differentiate $f(x) = \frac{\sin(2x)}{x}$.
$f'(x) = \frac{2\cos(2x) \cdot x - \sin(2x)}{x^2}$.

Chain rule with tables of values

10 Given $f(2)=5$, $f'(2)=3$, $g(2)=7$, $g'(2)=-1$, find $\frac{d}{dx}[f(g(x))]$ at $x=2$ if $g(2)=7$ and $f'(7)=4$ (note the composition uses f' evaluated at $g(2)=7$, not at 2).
$\frac{d}{dx}f(g(x))=f'(g(x))\cdot g'(x)$; at $x=2$: $f'(7)\cdot g'(2)=4\cdot(-1)=-4$.

Chain rule with inverse trig and logarithms

11 Differentiate $f(x)=\sin^{-1}(2x)$.
$f'(x)=\frac{2}{\sqrt{1-(2x)^2}}=\frac{2}{\sqrt{1-4x^2}}$.

12 Differentiate $f(x)=\ln(\sin x)$.
$f'(x)=\frac{\cos x}{\sin x}=\cot x$.

13 Differentiate $f(x)=\tan^{-1}(x^2)$.
$f'(x)=\frac{2x}{1+(x^2)^2}=\frac{2x}{1+x^4}$.

Finding tangent lines using the chain rule

14 Find the equation of the tangent line to $f(x)=(2x-1)^3$ at $x=1$.
$f(1)=1^3=1$. $f'(x)=3(2x-1)^2(2)=6(2x-1)^2$; $f'(1)=6(1)^2=6$. Tangent: $y-1=6(x-1)$, i.e. $y=6x-5$.

Visualizing a composite function and its derivative

15 The graph shows $f(x)=(x-1)^2$ (an inner-outer composite of $u=x-1$ and u^2), with its tangent line at $x=2$.

a Use the chain rule to find $f'(x)$. $f'(x)=2(x-1)(1)=2(x-1)$.

b Confirm $f'(2)$ matches the slope of the tangent line shown. $f'(2)=2(1)=2$, matching the tangent line $y=2x-2$, which has slope 2.

The chain rule in applied rate problems

16 The volume of a sphere is $V=\frac{4}{3}\pi r^3$, and the radius is growing according to $r(t)=2t+1$ (cm, seconds). Use the chain rule to find $\frac{dV}{dt}$ at $t=2$.
$\frac{dV}{dt}=\frac{dV}{dr}\cdot\frac{dr}{dt}=4\pi r^2\cdot 2$. At $t=2$: $r=5$, so $\frac{dV}{dt}=4\pi(25)(2)=200\pi\approx 628.3$ cm^3/s.

Chain rule with a table of composed values

17 Given $h(x) = f(g(x))$ where $g(1) = 3$, $g'(1) = 2$, $f(3) = 7$, $f'(3) = -4$, find $h'(1)$.
$h'(x) = f'(g(x)) \cdot g'(x)$: $h'(1) = f'(g(1)) \cdot g'(1) = f'(3) \cdot 2 = (-4)(2) = -8$.

18 Differentiate $f(x) = \sqrt[3]{x^2 + 1}$ using the chain rule.
$f(x) = (x^2 + 1)^{1/3}$: $f'(x) = \frac{1}{3}(x^2 + 1)^{-2/3}(2x) = \frac{2x}{3(x^2 + 1)^{2/3}}$.